Identifying a Confound Is Not Enough: Evaluating Debiasing Corrections in Drug Repurposing

Rishik Kondadadi[1] and John Ortega[2]

[1] University of Minnesota
[2] Northeastern University

Abstract. A model trained to rank antiepileptic drugs puts bemegride, a convulsant once used in electroconvulsive therapy, first among 1,522 non-antiepileptic candidates. It has learned to favor nervous-system drugs generally, well beyond epilepsy specifically. We test two fixes for this organ-system confound across five drug indications: a naive subtraction of a generic area score, and a more principled correction adapted from genomics (RUV-2) that estimates the confound from negative-control indications before removing it. Naive wins every bias metric we measure. That result does not settle the comparison, because a bias-diagnostic metric is itself a measurement with a detection limit. If it cannot see what the principled correction does, naive's win reflects the metric's blind spot, not the correction's quality. We built an alignment check for exactly this possibility, scoring each correction's own penalty term as a classifier of what the metric flags, then confirming the result with a direct intervention that dials the correction's real effect into the score. For three of five indications, the metric cannot detect the principled correction's effect at any strength we test; naive's advantage there is invisible, not earned. Epilepsy, the motivating case, is the exception. A real but smaller effect survives the check there. A simulation with a known ground-truth confound explains both outcomes. The principled method's estimated factor correlates with the true confound better than naive's own penalty does, 0.84 versus 0.68 AUROC, so it identifies the confound correctly. But the way it is applied, fit jointly with the base model and then dropped from the final score, fails to remove the confound from the ranking, while direct subtraction of that same factor removes it as effectively as naive subtraction does. The failure sits in the application step, not the identification step. That is a more general problem than simply being invisible to the bias metric.

1 Introduction

A plain ECFP4 fingerprint model, trained on known antiepileptic drugs, ranks bemegride first among 1,522 non-antiepileptic candidates. Bemegride is a convulsant once used in electroconvulsive therapy. We reproduce this exactly, score 0.7942, as this paper's motivating case; a companion study from this group

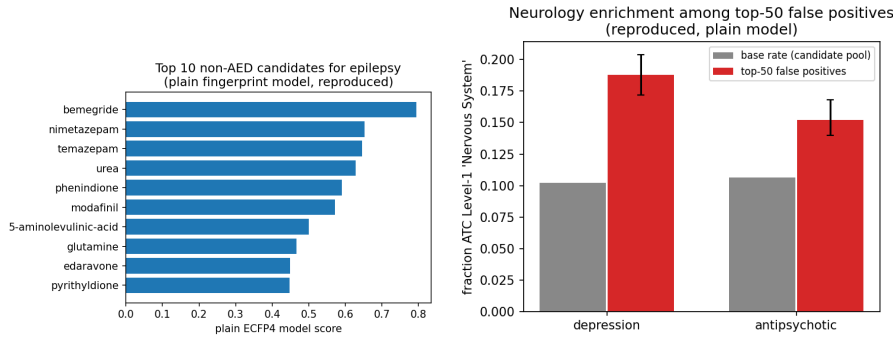


Fig. 1. Left: top 10 non-antiepileptic candidates ranked by a plain ECFP4 model; bemegride, a convulsant, ranks first. Right: nervous-system drugs (ATC 'N') enriched among top-50 false positives of depression/antipsychotic models vs. base rate.

audited evaluation-protocol choices in LINCS-based repurposing models and reported the same ranking [12], proposing no correction method. Chemical structure is not the only thing a repurposing model can learn to exploit. A model trained to separate antiepileptic drugs from everything else also learns, incidentally, what distinguishes neurology drugs from the rest of the pharmacopeia, and that second signal rides along with the first.

This case doubles as a puzzle about measurement. A bias-diagnostic metric is itself a measurement, and a measurement has a detection limit. Compare two debiasing corrections against one bias metric, and suppose one wins. The usual reading is that the winning correction removes the bias better. A second reading is available whenever the two corrections work through different mechanisms. The metric may simply be built to see one and not the other, in which case the comparison never tested what it looked like it was testing. Telling these readings apart means checking directly whether the metric could have detected the losing correction's effect at all, rather than inferring quality from the scoreboard alone. We call this an alignment check. It scores each correction's own penalty term, the quantity it subtracts or projects out, as a standalone classifier of whatever the metric flags. A term that classifies the flag near chance is a screening signal, not proof, that the metric cannot reward or penalize that mechanism; Section 5.6 tests this directly.

The same pattern appears in aggregate. Restricting to drugs with a WHO ATC classification, nervous-system drugs make up 18.8% of the top 50 false positives for a depression model and 15.2% for an antipsychotic model, against base rates of 10.2% and 10.6%, 1.5 to 1.8 times base rate, smaller than a name-matched Hub label had suggested; ATC covers only 9.7% of the LINCS library.

Two corrections suggest themselves. The simple one subtracts a generic area score from the disease-specific score, directly penalizing confounded-class membership; prior informal analysis assumed this collapses precision too far to use. The more principled one borrows an idea from genomics and pharmacoepidemiology: estimate the shared organ-system factor from negative-control indications, same area, different mechanism, and remove only that factor rather than a blunt

membership score. Built and evaluated, it did not beat the simple one on any bias-diagnostic metric measured, where this paper could have stopped and reported a negative result.

Instead we apply the alignment check, and it complicates the conclusion rather than resolving it. Section 5.5 shows the bias metrics' flags are predictable from the same base representation for three of five targets but not uniformly, and a direct intervention test (Section 5.6) confirms the metric shows no measurable response to the principled method there over the correction strengths tested, with a real, smaller exception for epilepsy. A synthetic experiment with a known confound (Section 5.6) explains why: the method's own factor recovers the true confound better than naive's penalty does, but how it is applied, fit jointly then projected out at scoring, does not reduce the confound's presence in the ranking the way naive's cruder subtraction does.

2 Related Work

Negative controls for unwanted variation. RUV-2 [8] estimates latent confounding factors from known-null control genes and removes them before testing. Later work [9] unifies several RUV variants. Surrogate variable analysis [16] and empirical-Bayes batch correction [11] address related but distinct versions; ComBat needs known batch labels, not negative controls. Negative-control exposures in pharmacoepidemiology [18] formalize the same logic for drug confounding [23]. Our area-factor step is a direct structural analogue: indications rather than genes.

Debiasing drug-repurposing and DTI models. NRBdMF [1] factorizes a signed drug×disease matrix from SIDER [14] to reduce false positives through within-drug directionality. It never uses other diseases as negative controls. TAPB [17] and DRDM [19] address other biases, target-popularity and graph-degree. Neither targets organ-system confounding. Campillos et al. [3,13] first showed SIDER-derived phenotypic similarity carries repositioning signal; that result is the basis for our hard negatives.

Benchmarks and resources. repoDB [2] provides a curated gold-standard set of approved/failed drug-disease pairs; we use it as a standard secondary comparison point. The Drug Repurposing Hub [5], a name-matched label used in prior work, has limited coverage. We replace it with ATC classification. Epileptogenic/proconvulsant lists have an established clinical literature base [20,4,21]. Two lines of work bear on evaluation validity: scaffold splits overestimate performance relative to a realistic split [10], and affinity benchmarks reward memorization unless reweighted [25]. Neither asks whether a bias-diagnostic metric can see a correction's mechanism at all.

3 Data

We build on the nine ChEMBL [24] indication classes used in our companion study (full list in the repository), over the same 9,878-compound LINCS library, extended with nine further ATC-defined indications so every original class has at least one same-area, different-mechanism negative-control partner. ATC Level-1 codes come from the ChEMBL molecule API by full InChIKey; only 954 of 9,878 compounds (9.7%) carry any ATC classification, why Section 5.1's enrichment is smaller than a name-matched Hub label had suggested.

SIDER [14] side-effect data is joined by PubChem CID, extending a prior 1,725-compound join; coverage remains low, 543 of 9,878 (5.5%), for the same reason as ATC. We use its adverse-effect terms as hard negatives (convulsion-family for epilepsy, hypotension for antihypertensives, hyperglycaemia for antidiabetics) where such a term map makes sense, not invented otherwise.

For epilepsy, we use a 43-drug proconvulsant list from this group's own prior, unpublished work, stated plainly rather than as an external resource: literature-curated from Chen, Albertson & Olson [4], a peer-reviewed review of drug-induced seizures, cross-validated against two public sources. Of 82 canonical compounds, 62 resolve to our library; 38 (61%) carry a SIDER-listed convulsion/seizure effect, and 24 (39%) match a risk tier in Ruffmann, Bogliun & Beghi's independent review [20].

4 Method

Each compound is represented by concatenating two pieces. The first is a 32-component PCA of its ECFP4 fingerprint (TruncatedSVD, no indication-matrix leakage). The second is a 16-dim embedding from a free logistic matrix factorization fit jointly across all 18 indications; its gradient is masked to training cells only within each fold, so held-out labels never enter the embedding. Chemistry is necessary, not cosmetic. A pure co-occurrence factorization carries almost no signal here. 94% of the library has zero positive indications, which is why NRLMF/NRBdMF add drug or disease similarity, and why we add chemistry for the same reason. Under CVS2, a held-out compound's row is masked for every column, so its MF embedding stays at random initialization. That injects noise into the logit at roughly 5–10% of chemistry's magnitude. Dropping the MF embedding and rerunning CVS2 on chemistry alone (code/24_chemistry_only_ablation.py) reproduces Tables 1–2 within 0.02, and epilepsy's Table 3 row within 0.001. Depression's and antipsychotic's rows shift up 0.03–0.07 without this noise, but stay well below naive's 0.60–0.79, so no claim below depends on it.

On top of this representation we compare four scorings for target indication k_0. *Plain* fits one logistic regression per indication, no area correction. *Naive subtraction* scores the plain score minus λ times a min-max-normalized mean indication rate over k_0's negative-control set, $\lambda = 2$. *NRBdMF* factorizes a signed drug-by-disease matrix [1]: -1 for a ChEMBL/ATC indication, $+1$ for a SIDER

adverse-effect hit. Chemistry is withheld, matching the original specification. *Proposed* starts from k_0's negative-control set $\mathrm{NC}(k_0)$, same ATC area, different mechanism. It takes the train-only residuals of the already-fit plain models on $\mathrm{NC}(k_0)$'s columns, and uses their leading left singular vectors as an area factor A (Gagnon-Bartsch and Speed's RUV-2 strategy [8], applied to indications rather than genes). It then refits k_0 with [chemistry+MF, A] and scores with A's coefficients dropped.

We later tested a shrunk version of this correction, scaling A by $s \in [0, 1]$ following Gerard and Stephens' unified RUV family [9]. Section 5.4 reports no improvement.

4.1 The alignment check

Any correction scoring a candidate as a base score minus (or plus) a penalty term can be probed independently of whatever metric judges it. Naive subtraction's λ-weighted area rate and the proposed method's projected-out A-coefficients both have this form. Three scores matter for the proposed method: $s_{\text{plain}} = \hat{\beta}_0^\top x$ (fit without A), $s_{\text{joint}} = \hat{\beta}_X^\top x + \hat{\gamma}^\top a$ (refit with A), and $s_{\text{proposed}} = \hat{\beta}_X^\top x$ (A dropped). $P = s_{\text{joint}} - s_{\text{proposed}} = \hat{\gamma}^\top a$ is what this check and Section 5.6 score, not $s_{\text{plain}} - s_{\text{proposed}}$. Refitting with A also moves the base coefficients: $\mathrm{corr}(s_{\text{plain}}, \sigma(\hat{\beta}_X^\top x)) = 0.96$–$1.00$, a coefficient shift of 2–12% across targets (code/19_intervention_doseresponse.py). The two are close here, not identical. Their top-50 rankings overlap 84–98% with rank correlation 0.98–1.00, narrowest for epilepsy, the target whose base coefficients shift most. A bias-diagnostic metric is defined through a binary flag F: same-area membership, hard-negative membership, or any other criterion the metric penalizes. The check scores the correction's own penalty term P as a standalone classifier of F, via $\mathrm{AUROC}(P, F)$. Near-0.5 AUROC means P carries little marginal rank information about F in isolation. That makes it a heuristic screening diagnostic, not a formal necessary condition, since an interaction with the base score could still move a ranking metric without a marginal association. Section 5.6 tests that directly. Short of it, a comparison against a baseline whose term *is* informative about F is not on equal footing. We apply this to our two corrections in Section 5.5.

Section 5.5's result has a structural reason. **Proposition.** Let X be the shared representation on which every negative-control indication's model is fit. The fit uses L_2-penalized logistic regression at inverse-strength C (weights W), giving coefficients $\hat{\beta}_j$ and fitted probabilities $\hat{p}_j$ on train rows only. Each residual $r_j = y_j - \hat{p}_j$ satisfies $X_{\text{tr}}^\top W r_j = \hat{\beta}_j / C$. Orthogonality to X is exact only as $C \to \infty$; at finite C the deviation is proportional to $\hat{\beta}_j$, not zero. A, the leading left singular vectors of $R = [r_1, \ldots, r_{|\mathrm{NC}|}]$, inherits whatever orthogonality R has.

We measured this rather than assuming it away. At the $C = 1$ used throughout, A's projection onto X's column space holds 34–40% of A's variance, not

Table 1. AUPRC on true indications, CVS2 (mean over 5 reruns; 95% bootstrap CI in parentheses).

Target	Plain	Naive subtract	NRBdMF	Proposed
depression	0.201 (.114,.289)	0.190 (.105,.262)	0.039	0.181 (.112,.248)
antipsychotic	0.440 (.252,.628)	0.412 (.215,.608)	0.039	0.438 (.264,.612)
epilepsy	0.091 (.065,.127)	0.090 (.067,.128)	0.024	0.085 (.065,.112)

the near-zero a pure guarantee would give. Section 5.5's unregularized OLS fit is where the exact case holds. Section 5.5 tests both halves empirically.

5 Experiments

5.1 Reproducing the motivating confound

We retrained the plain ECFP4 model for epilepsy, depression, and antipsychotics, averaged over five seeds. Both parts of Section 1's motivating observation reproduce exactly: bemegride ranks first of 1,522 non-antiepileptic candidates, and the same 1.5–1.8x nervous-system enrichment holds for depression and antipsychotic. We could not locate, in any saved script, the AUPRC collapse from 0.45 to 0.17 that an earlier analysis attributed to naive subtraction. We report what we can measure.

5.2 Evaluation protocol

We evaluate under three cross-validation schemes: CVS1 (random 20% cell hold-out), CVS2 (cold-start-drug, 20% of compounds), and CVS3 (cold-start-disease, the entire target column). Each is run five times on an identical 1,702-compound pool shared by all four methods.

5.3 Main results

Table 1 reports CVS2 AUPRC on the true indication label; CVS1 and CVS3 show the same pattern and are in the repository. Table 2 reports the two bias-diagnostic metrics that matter most here: the same-ATC-area rate among the top 50 false positives, and the median rank percentile of SIDER-flagged hard negatives.

NRBdMF performs close to chance everywhere, AUPRC 0.02 to 0.15. A pure within-drug signed factorization, with no chemistry, has little to work with on an 18-indication matrix this sparse. The proposed method tracks plain closely on AUPRC, the precision-preservation property it was designed for. But on the depression/antipsychotic/epilepsy bias metrics in Table 2, it is dominated by naive subtraction. The clearest case is the proconvulsant panel, where naive pushes it from the 31st percentile to the 72nd while proposed moves it under 5 points, and in the epilepsy CVS2 row moves it in the wrong direction. Naive's AUPRC cost is modest across every row and CVS scheme, never approaching

Table 2. Bias-diagnostic metrics. Top-50 same-area rate: lower is better (base rate ≈ 0.09–0.11). Hard-negative/proconvulsant rank: higher percentile is better.

Metric / Target	CVS	Plain	Naive subtract	NRBdMF	Proposed
area rate, depression	2	0.112	0.024	0.084	0.112
area rate, antipsychotic	2	0.136	0.096	0.084	0.144
area rate, epilepsy	2	0.144	0.068	0.120	0.180
hard-neg. pct., epilepsy	2	38.4	50.9	47.8	41.9
proconv. pct., epilepsy	2	30.9	71.8	49.2	26.4

the collapse an earlier informal analysis had reported (Section 5.1). Betablocker shows no method differentiation, area rate 0.108 to 0.112. Lipid reproduces the core pattern: naive halves it, 0.104 to 0.052, at no AUPRC cost, while proposed moves neither. A reconstructed DRDM baseline behaves like plain throughout, consistent with targeting a different bias (Section 2).

5.4 Robustness, validity, and identifiability follow-ups

A method that identifies an area factor from negative controls should leave each control's predictability roughly unchanged when corrected using a factor from the *other* controls, if those controls are mechanistically independent. We test this with a leave-one-negative-control-out diagnostic over every (target, control) pair. 13 of 14 cells cluster within ±0.03 of zero AUPRC change, consistent with independence. The exception, anxiolytic, is small-sample noise from its 7-positive class: 20 seeds shrink its mean delta to +0.034, within 1 SE of zero. We tested two further candidates and set both aside. First, the negative-control classes are small, 4 to 21 positives each, so the area factor could be mostly noise. Empirical-Bayes-style shrinkage, $A \leftarrow sA$ for $s \in \{0, 0.25, 0.5, 0.75, 1.0\}$ selected via the same diagnostic, picked the full-strength, unshrunk correction ($s = 1$) for all three targets. Second, RUV's zero-loading assumption [9] could fail, letting the area factor absorb target signal along with the confound. Regressing the true label (train rows) onto the fitted area factor gives R^2 0.0006–0.021 for all six cells, which does not track naive's advantage; epilepsy has the smallest overlap but the largest gap in naive's favor.

5.5 Applying the alignment check

Finding no identifiability violation leaves a more mundane candidate. The bias metrics may simply sit closer to naive's raw negative-control membership than to proposed's residualized factor. We apply the alignment check (Section 4.1) to test this. For each target, under CVS2, we compute naive's penalty term and proposed's term, the contribution A's coefficients would have made had they not been dropped, for every non-indicated test compound. We score each as a classifier of the exact compounds the bias metrics flag: same-ATC-area, SIDER-hard-negative, or proconvulsant-panel membership.

The SVD's own output for a held-out row is exactly zero by construction: a zero residual row, forced by the negative controls' own train mask, $\sim 10^{-16}$.

Table 3. AUROC of each quantity vs. the flagged compounds (mean, 5 seeds, CVS2). 0.5 = no information. P and Plain→Proposed are made sign-free per seed before averaging across seeds, not after (see text); Plain→Proposed is the full empirical plain-vs-proposed gap.

Target	Flag	Naive	P	Plain→Proposed	X alone
depression	same-ATC-area	0.766	0.561	0.575	0.718
depression	hard-negative	0.636	0.540	0.556	0.514
antipsychotic	same-ATC-area	0.774	0.579	0.568	0.706
antipsychotic	hard-negative	0.663	0.580	0.568	0.591
epilepsy	same-ATC-area	0.787	0.657	0.653	0.722
epilepsy	hard-negative	0.600	0.568	0.559	0.613
epilepsy	proconvulsant	0.721	0.676	0.677	0.695

Scoring P directly from it would give every test compound the same, uninformative value regardless of its true factor. We instead fit a linear out-of-sample projection from X to the train rows' own factor values (5-fold CV R^2 0.44–0.56, non-degenerate), and use that projection for test rows below.

Naive's term separates every flag well above chance, 0.60 to 0.79, regardless of how P is computed below. P's sign is arbitrary within each seed, since the SVD that produces A fixes a sign only up to that ambiguity. Aggregating $\max(\text{AUROC}, 1-\text{AUROC})$ of the AUROC already averaged across seeds, rather than of each seed's own AUROC before averaging, lets a real per-seed effect cancel toward 0.5 whenever the sign lands differently across independent refits. That happens exactly in depression's and antipsychotic's cells; epilepsy's sign is stable across seeds. Made sign-free per seed before averaging, the convention code/23_permutation_check.py already uses, P is 0.54–0.58 for depression and antipsychotic. That is modest rather than literally at chance, but still well below naive's 0.64–0.77. Plain→Proposed, the full empirical plain-to-proposed gap, is comparable to P on most cells, 0.53–0.68 both, though the two are related, not interchangeable (Section 4.1). Against the undropped joint fit instead of proposed, that same gap is lower on 6 of 7 cells (results/score_decomposition_alignment.csv); the two effects partially cancel rather than add when combined into one score. $\max(\text{AUROC}, 1-\text{AUROC})$ is itself upward-biased under the null when its sign is picked on the same evaluation set it is reported on. We ran a permutation test: shuffle the flag within each seed's split, 5000 times (code/23_permutation_check.py), combined across seeds with Fisher's method against a Bonferroni threshold for the 7 cells ($\alpha = 0.0071$). Depression stays non-significant ($p = 0.12, 0.66$). Antipsychotic's two flags and epilepsy's hard-negative pass on all 5 seeds ($p =$ 0.0028–0.024), but collapse ($p \geq 0.18$) with the single most extreme seed dropped, a seed-specific artifact. Epilepsy's same-ATC-area holds up either way ($p = 2.2 \times 10^{-9}$ all 5 seeds, $p = 5.4 \times 10^{-7}$ excluding the most extreme). Proconvulsant clears the raw threshold ($p = 0.0004$) but not α once that seed is dropped ($p = 0.01$). Fisher's method also assumes the 5 seeds are independent. They are not, since all 5 re-split one compound pool. A compound-level permutation modeling that directly (code/25_hierarchical_permutation.py) agrees with Fisher except on proconvulsant, where it finds significance ($p = 0.0027$) that the leave-one-out check does not. That result is real under the test built for this concern, but fragile under the other. By the Proposition, proposed's term

is only partially orthogonal, on train rows, to X. The rightmost column tests, rather than assumes, whether F is predictable from X. Same-ATC-area is well predicted by X on every target, 0.706 to 0.722, close to naive's own 0.766 to 0.787. Most of what marks a compound as belonging to the confounded area is already present in X before any negative-control information is added. This does not explain why epilepsy's area term specifically carries signal while depression's and antipsychotic's do not. Section 5.6 does.

This two-step recipe names nothing drug-repurposing-specific; it diagnoses association, not what the metric does when the real effect is applied, which Section 5.6 tests directly.

We also re-ran this check outside drug repurposing, on two genomics datasets with a known ground-truth confound: Vawter et al.'s brain gene-expression study [22] (gender vs. lab, 799 housekeeping controls [7]), and bladderbatch [6,15] (3-level cancer status vs. 5-level batch). Step one holds exactly in both: each control gene's OLS residual satisfies the normal equations to floating-point precision. Step two diverges. Lab is unpredictable from gender (AUROC 0.39–0.41), yet the orthogonalized factor stays informative about it, 0.50–0.94 alone and 0.95–1.00 combined, beating naive's 0.55–0.76. Batch is moderately predictable from cancer status (0.722), yet the orthogonalized factor stays as informative about batch as naive's own (0.68 vs. 0.62). The check's second step turns on predictability of F from X alone, not on whether F and the covariate of interest are related.

5.6 Confirming it directly: intervention, and a synthetic ground truth

Standalone AUROC does not show whether Table 2's metric actually moves when a penalty's real effect is dosed into the score. We sweep how much of each penalty is included, $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$, where $\alpha = 1$ for naive and $\alpha = 0$ for proposed reproduce Table 2. We recompute the bias metrics at each dose, same CVS2 split, all five targets. Naive's metrics move strongly and monotonically for depression, antipsychotic, and lipid; epilepsy's proconvulsant percentile goes from 30.9 at $\alpha = 0$ to 71.8 at $\alpha = 1$. Proposed's metrics stay exactly flat for the same three, depression area rate 0.112 and lipid 0.104 throughout. Betablocker moves little under either. Epilepsy is again the exception: proposed's metrics move a small real amount, area rate 0.180 to 0.140 and proconvulsant percentile 26.4 to 32.7, a quarter of naive's swing. This confirms the alignment claim operationally for three of five targets, the same exception as Table 3. Unlike standalone AUROC, this measurement is not subject to Section 5.5's sign-selection caveat, so epilepsy's small shift stands as independent evidence regardless.

A real-data flag is a proxy; the true confound is unobserved. A simulation fixes that. It uses 3,000 compounds, chemistry X, and an organ-area confound z noisily tied to a chemistry subspace. Indication log-odds combine a disease-specific term with, for same-area diseases, z; negative controls share the area with an independently drawn mechanism. We run 30 independent seeds, each redrawing X, loadings, and labels. AUPRC (0.484/0.273/0.485 plain/naive/proposed) reproduces Table 1's pattern. Figure 2a compares five scores on the top-50 rate

of *true* high-confound compounds, base rate 0.29. Naive substantially reduces it; proposed does not. Direct subtraction of proposed's own estimated factor, the identical estimate applied the other way, reduces it as much as naive does: a paired gap of 0.218 (95% CI 0.191–0.245, $d_z = 3.0$, $p < 10^{-15}$) attributable to application mechanism alone. Proposed's factor AUROC against the true confound is 0.84, naive's own 0.68. It identifies the confound better, but projecting it out at scoring does not act on what it identified. Figure 2b sweeps estimation quality directly, the same two mechanisms on the same noisy estimate at seven levels. The gap is near zero at the noise-free extreme, peaks at moderate noise near the method's own 0.84-AUROC quality, and narrows again at high noise as both converge to the uncorrected baseline: non-monotonic rather than growing without bound. Where the factor carries little real signal, depression, antipsychotic, lipid, there is little for this gap to act on. Where it does, epilepsy, the gap becomes the effect measured here.

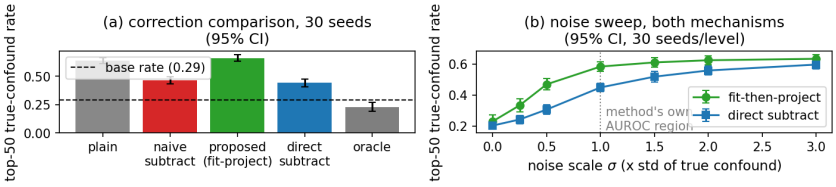


Fig. 2. Synthetic experiment, 30 seeds, 95% CI. (a) Top-50 true-confound rate by score. (b) Both application mechanisms over seven noise levels on the identical estimate.

This gap holds across a tenfold range of confound-chemistry correlation (code/28_confound_chemistry_correlation_sensitivity.py, $R^2(z, X)$ 0.07–0.80, paired gap 0.03–0.26). It vanishes only where X already explains z almost entirely ($R^2 \geq 0.94$). That makes it specific to confounds the representation does not already absorb, not an artifact of one fixed structural choice.

6 Discussion and Limitations

This paper set out to build a more principled alternative to a correction assumed too costly to use. Naive won every bias metric measured. Whether that scoreboard meant what it appeared to is target-dependent. For depression, antipsychotic, and lipid, the metric showed no measurable response to the proposed method's mechanism over the correction strengths tested. For epilepsy's area and proconvulsant flags, it partially did (Section 5.6). Winning a bias metric means suppressing a flagged signal, not a validated improvement. Organ-system overlap can itself be real pharmacology, so naive's larger reduction may be only a stronger diagnostic response, not better candidates.

The check and the correction-application gap are the more durable contributions. The residual-orthogonality result (Proposition) follows from standard regression score equations; the contribution here is the diagnostic framework and the confirming experiments, not new theory. Reproducing P (Section 5.5) depends on two computational details. It requires an out-of-sample projection

rather than the SVD's own held-out output, which is zero by construction. Its sign must also be fixed per seed before averaging. DRDM [19] has no public code, so its baseline here is a reconstruction, not a verified one.

Limitations remain. ATC/SIDER cover under 10% of the library. CVS3 collapses plain and proposed. repoDB [2] overlap is too little for an AUPRC check. Code: `https://github.com/edanmn/area-debiasing-drug-repurposing`.

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